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Technology Integration in a Learning Environment

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Technology Integration in a Learning Environment: Formulating a Researchable Question

What is the most effective functional integration of the basic and optional features of software packages (Web Advisor and its relational database, Colleague) currently in use by Ohlone Students, Faculty, and Staff?

Reviewing the Literature and Obtaining the Latest Information on the Subject

In today's "Information Age," technology is a crucial tool in delivering goods and services, including those offered by higher education. As the Internet and its related technologies have revolutionized world commerce and communication, they have also greatly affected the ways colleges and universities operate. In both cases, the key component is providing better communication between stakeholders.

Many colleges and universities are using the same kind of technology for teaching and learning and communication as are used by corporations. Like many other institutions of higher education, Ohlone College has implemented the software application Web Advisor by DataTel, a Web-based communication application that allows administrator/employee, faculty, and student access to information contained in the administrative database. This application offers a diverse array of functions, including allowing a school's administrator to track student enrollment, faculty and student to track class rosters, reporting of grades, as well as a grid for communication between all parties. In addition to these functions, Web Advisor offers many others that have not yet been utilized by our college. Currently, Ohlone has implemented only the most basic functions of Web Advisor, creating some areas in which the software is less than "user friendly."

Moreover, since Web Advisor is just one of the many software applications supporting

the teaching and learning at Ohlone (ex. email and other communication functions), many loosely connected functionalities have manifested without full integration into the larger system. The ideal application would be a comprehensive software solution that supports all the teaching and learning and collaboration needs. The scope of this investigation is a proposal of the Internet portal with all the necessary knowledge-based, technology requirements, and Web Advisor. One example of such technologies is the Internet portal application for student, faculty, and employee. Web Advisor is a Web service of the Internet portal. Technology requirements are the software packages, operating systems, databases, and physical hardware devices (i.e. the physical infrastructure). Although most Internet portal users are aware of the physical- or human-support infrastructure, the reality is that the technology infrastructure makes it all possible. The physical infrastructure is a backbone to deliver Web Advisor and the delivery of the Internet portal. All of these investments are building to support the teaching and learning needs.

The Internet portal cannot exist without the physical infrastructure, and is a one component of the technology infrastructure. The physical infrastructure includes the Web server, the database server, computer machines, hardware devices, operating software, and Web Advisor. Depending on the type of computer machine in use, it installs with the appropriate hardware devices and operating software. A computer machine can be either a client or a server machine, and is networked to one or more servers on the network. All computer machines connected together in this way are known to be a part of the same network. In addition, there is a physical network (i.e., wires, routers, bridges, and fiber and Ethernet). A computer machine operates by the operating software that enables it to work (ex. Internet Browser, Windows XP, Microsoft Word, etc). For any computer machine or server to work, operating software needs to

be installed. In addition, the infrastructure can include telecommunications links to the world outside the campus, and between campuses, including telephone services, videoconferencing equipment and networks, and the Internet.

Of course, no physical infrastructure can function without human support as well. In fact, in many ways it can be considered even more important than the physical infrastructure. It is the people who not only keep things running and up-to-date, but who also provide help and instruction to users who must learn to operate the hardware and software. There are four levels of such human support. The first support level is the technology infrastructure support staff; they are the technical support people who ensure that the network and equipment are working properly, including installation, operation, and maintenance. The second support level is the educational technology support staff; they are the media production and services staff (e.g., interface designers, graphics designers, and etc.). The third support level is the instructional design staff and they are the education services people (e.g., instructional design, faculty development, project management, and evaluation). The fourth support level are the subject experts (e.g., professors, and instructors). Both the technology support staff and the educational technology support staff are responsible for operational aspects. In contrast to the instructional design staff and the subject experts, they are the people responsible for the creation and delivery technology-based teaching. Teaching and learning through technology is only possible if the human support staff makes it available and enables information to be delivered.

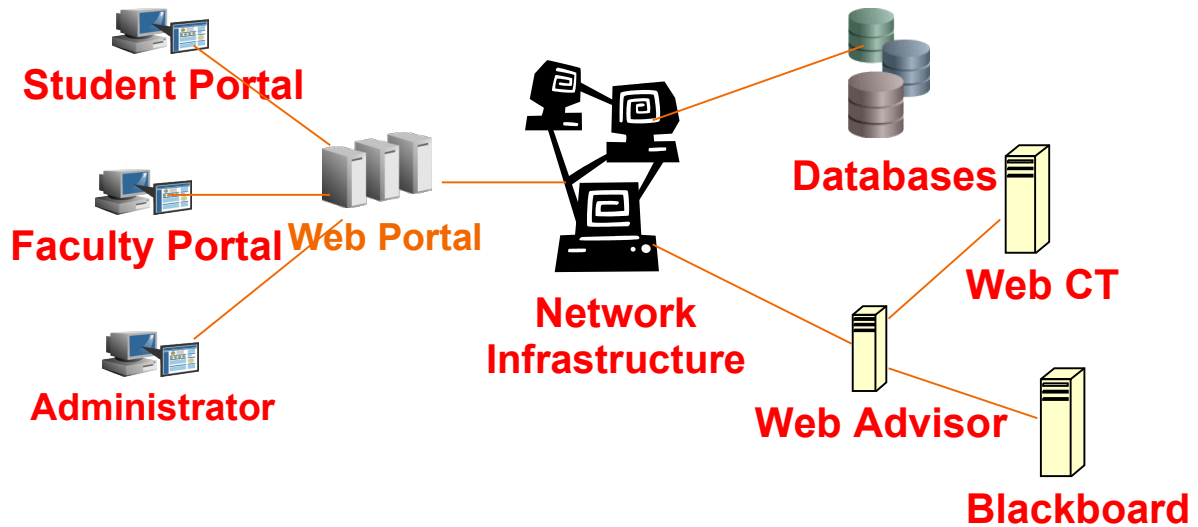
Web Advisor Functionalities

This table shows a list of applications available in Web Advisor. A comprehensive technology solution for teaching and learning includes a student, faculty, and administrator portal that contains Web Advisor, email, portfolio, BlackBoard, WebCT, and a limited student advisor

and student services module.

Self-Service for Students		Self-Service for Faculty	
Change password	active	Change password	active
Login request	not used	Login request	not used
	active- includes search engine		
Course Availability	for classes	Faculty Schedule	active
Admissions status	not used	Advisee List	active
			active- includes PA hours submission and no-
FA by Year	active	Intermediate and Final Grading	show/withdrawal indication
			active-
FA by Term	active	Class Roster	customized to show student contact information
Cumulative GPA	active	Position Summary	inactive
GPA by Term	active	Leave Plan Summary	inactive
		Faculty Stipend Summary	inactive
Transcript	active		Web Advisor and BlackBoard
		BlackBoard	Integration
Registration	active		Web Advisor and WebCT
		Web CT	Integration
Class Schedule	active		Web Advisor and Webmail
Grades - Midterm and Final	active	Email	Integration
			Student Services
Make a Payment	active	Student Services	Integration
			Student
Account Summary	active	ePortfolio	ePortfolio
	active- includes link to counseling		
Test Score Summary	used by FA	Faculty Advisor	Faculty Advisor
Missing and Received Documents			Integration
	active for evaluators,		
Program Evaluation			

	counselors
Transcript Request (with e-commerce)	in process
Graduation Application	active
Address Change	active
Financial Aid Award	
Letter Online	active
FA Loan Application	not used
	Web Advisor and BlackBoard
BlackBoard	Integration Web Advisor and WebCT
Web CT	Integration Web Advisor and Webmail
Email	Integration Student
ePortfolio	ePortfolio Student Services
Student Services	Integration Student Advisor
Student Advisor	Integration

Technology Infrastructure for Web Advisor

Preparing an In-depth Response to the Research Question

In the sweeping momentum of the virtual university, Ohlone College, as other colleges and universities, finds itself in need of maximizing its technology infrastructure, while taking the best advantage of upgrades to better serve students and to maintain coherence in faculty uses of course management software. There is very much a micro-level need being addressed: that of keeping students and creating a techno-culture appropriate to the abilities and needs of the faculty and staff that make up the college collective. But there is also a macro-level need. The college is, in this proposed project, acknowledging the “new mode of production” in the face of “the emerging and applied user-shaped kind of knowledge that is now required” in the modern global society (Robins, 2002 p. 42). Citizens in this view are increasingly accustomed to, expecting or actively learning about how to obtain information for themselves. “New undergraduates increasingly want to complete registration procedures online rather than wait in line at a desk, and so on” (Robins, 2002 101). Ease of student access to information is but one

strand of the triple helix that contains students, faculty and staff but serves as a starting point.

In the global context, colleges like Ohlone College function as a “site of interconnectivity” in the “knowledge society” (Robins, 2002). As such the institution is being “forced to be more reflexive in its relationship to its environment” and to “be responsible to multiple social forces rather than an autonomous site of learning” (Robins, 2002 p. 45). This, too, is the context within which Ohlone is considering these proposed uses of technology.

It seems also that the “actor network” approach is useful here to conceptualize the “network building” we are attempting (Robins, 2002 p. 91). When “central actors are treated not simply as scientists or technologists” but as those who are innovators, true collaboration occurs, and the knowledge base increases. In addition, and perhaps most importantly, in this approach “agency is not restricted to human beings but is expanded to encompass objects, machines and texts, all of which must also be enrolled and aligned” (Robins, 2002 p. 91). This reduces the tendency to reductionistic and dismissive “man versus machine” thinking and enables the visions contained in this project.

When students increasingly interface on a “one-to-one basis, but mediated through the technologies” (Robins, 2002 p. 92). Students also are increasingly embracing (by design or default) the ala carte concept of education, or a “mix and match” of their educations between institutions (Robins, 2002). This is also testimony to the impetus that has brought together student services and faculty needs in such a pivotal way.

Further, in the face of market mentality and the commodification of education, streamlining our online deliveries is an institutionally necessary move to ensure even our own “relevance” in some views. “There will be an increased regulation of universities, somewhat comparable to that experienced by many other industries and occupations. Such regulation is

greatly enhanced by new modes of information gathering that enable individuals, departments and institutions to be compared with each other, often across national borders” (Robins, 2002 p. 26). While we might not like to think of ourselves in this way, it’s increasingly true, that students are not tied to a single institution for the duration of their education, whether degree seeking, or for continuing education.

In this “audit society” colleges become “more transparent” and “accountable” organizations (Robins, 2002) even, it would seem tied to accreditation on the level of student learning outcomes (SLO’s). We do, however, want to be clear that we are not compromising our mission(s), and hope that in our design, we can call upon the “‘forces pushing democratization’ and weaken the ‘forces pushing’ commodification” (Robins, 2002 p. 201).

In this effort we are attempting to begin with the end in mind, which has been difficult in the face of “the speed of change [that] defeats broad comprehensive planning” (Robins 2002, p. 302). Nonetheless, we would do well to remember Francis Bacon’s words: “It is not possible to run a course aright when the goal itself is not rightly placed” (Robins, 2002 p. 173). It is our hope that this document can prove us to “be of use to policy makers” (Robins, 2002 p. 301) and reside wholly neither in the daily world of the classroom, or boardroom, nor somehow in the conceptual stratosphere of the learned “elite,” somewhere between being and doing, producing and consuming, being useful and “useless” (Robins, 2002), pragmatic and static. To not fall prey to the fallacy of generalization (Robins, 2002 p. 261) and bandwagon we will here attempt to “approximate empirical reality” (Robins, 2002 p. 302) to the best of our (limited) abilities within the scope of this final project.

Critically Assessing and Evaluating Key Aspects of the Available Information

At least as important as the creation and implementation of new technology in higher education is considering how such technology will be welcomed (or not welcomed) by the stakeholders involved. For this reason, it is imperative that Ohlone College creates a plan for introducing any changes in Web Advisor and its related tools to Students, Staff, and Faculty. Research reflects a trend toward successful implementation when certain factors are taken into account and openly addressed. One of these factors is the recognition that technologies are not entities that exist outside the campus culture. As Gopalakrishnan (2006) reminds us, “technology integration is not simply a process of putting proven knowledge into practice. Its ultimate goal is to make technology not an afterthought but a part of the teacher's everyday thinking” (p. 5). In other words, successful integration of a new technology involves encouraging its seamless assimilation into the larger functions of the stakeholders.

Some of the factors that have been shown to best encourage the regular use of new technologies in education include the following:

1. Technology support must be available on demand.
2. It must provide personal encouragement, instructional mentoring, and technical support of a routine nature.
3. Support must not necessarily come from Administrative-level technical staff, rather it can come from educational and technical orientations as long as they are able to "translate" between the two domains and work with users of varying technical abilities.
4. Program leaders need to be aware of the possibility that they may be creating support roles that did not exist before, and be sensitive to the fact that support staff are being asked to operate in a leadership role without requisite authority.

5. As a result, program leaders must publicly legitimize the roles of these support persons, and encourage stakeholders to cooperate with them and be receptive to their ideas.

(Gopalakrishnan, 2006).

As reported by Gopalakrishnan (2006), one of the pitfalls of technology implementation in many schools is the lack of requisite authority to “force” implementation by staff and faculty. As stated above, the author’s suggested remedy is to empower these change agents with authority. Some research (Cialdini, 1998) suggests that, rather than empowerment through perceived authority, a subtler, more psychological approach is often more successful. Therefore, insofar as implementation of the changes proposed in this document, the following program of integration is advisable. First, based upon the findings of Cialdini (1998), it is recognized that persuasive messages are most effective in changing behavior when delivered by a “like other.” Such factors as age, and gender of the participants may be of some use in this regard, for instance. The most significant factor in the organization of campus culture of Ohlone College, however, is usually two-fold, namely the department/division in which the individual works, and the number of years the individual has worked at the college. If a training program were developed in which members of various departments/divisions, and members of several “hiring cohorts” were trained to utilize the new technologies being implemented, and if those parties were offered incentives (e.g., Faculty or Staff Flex credit) for training “like others,” messages encouraging the use of the new technologies would likely be more positively received.

This program of implementation could likely serve as a pilot program for encouraging the use of several new technologies being proposed for the future of Ohlone College. One of these, “Podcasting” shows particular promise, allowing Faculty to digitally record, store, and deliver lectures and other information to their students, as well as to Faculty and Staff. The introduction

of such a technology will likely prove more difficult than that proposed in this paper, however, due to the fact that it is a relatively new idea for many in the field of education (Eash, 2006), as well as to the possibility that it may be perceived as an “intrusion” into mainstream teaching modalities. For these and other reasons, careful consideration of how best to present the proposed changes in technology must be made to ensure a smooth transition for all parties.

Formulating a Professional Proposal for Improvement

As mentioned earlier, the basic and optional features of software packages currently in use by Ohlone Students, Faculty, and Staff include at least three district tools: Web Advisor and its relational database, Colleague, is used by the college as the overall archive of student, financial and employee records. It supports active registration, as well as cashiering, advising and evaluation functions. It also provides faculty with real time enrollment information. The course management system - used primarily by faculty and students - functions to support virtual communication and collaboration that is course specific.

This software package is currently “passively linked” to the DataTel system in that it regularly adds newly enrolled students to the faculty class roster; however, it does not remove students from the roster, or mark them as dropped. The students, even if dropped, maintain access to the course all the features of the course management software and unless manually deleted by the instructor, have access to all materials posted, can take part in the online discussion etc. This also includes the student in any emails the instructor chooses to send to the entire class. This access remains active for an entire term, at which point all information is deleted. The course management system also allows for multiple grade entry and final grade entry, however this information is not transferred to the DataTel system and must be re-entered by the faculty to actually post grades for a student. This creates a dual, manual data-entry tats

for faculty.

The course management software allows for a robust email communication system, allowing students to contact faculty through the system this allows the faculty, as well as the students to direct their email to whatever email account they choose, allows the faculty and students to communicate with a whole class or individuals via email and provides for a variety of discussion boards and file sharing opportunities. The campus email system provides every faculty with an email, and requests that every student provide his or her own email information. This information, while available within the relational database, requires data retrieval skills to actually use.

What is being considered is the possibility of linking the data stored by, displayed by and entered across at least two of these systems - DataTel and Course Management software. Also suggested is the placement of these tools into a single log-on and web site of some kind to allow for more streamlined access to both students and faculty viewing of these records. While students would be able to view all of their courses, and their academic and financial records, faculty would have the ability to view only that information related to the students' enrollment in their courses. If the student chooses a specific Faculty to be his or her advisor, this information would become available to that Faculty member. What is being suggested is to combine these services into two major areas. One would be **Virtual Student Services**, which would allow the student to access almost all aspects of doing business with the college, and to maintain his or her student record. The other would be **Virtual Classroom Management**, which would allow Faculty a tool to coordinate and control a virtual (or physical) classroom, communicate with students, and maintain their own course records.

Virtual Student Service (WebAdvisor)	Virtual Course Management (Blackboard, WEBCT)	Linked Systems
Application and Admission	Post class information	One space/One logon for student faculty
Orientation	Communicate with individuals and/or whole classes	Virtual Classroom tools for all sections
Course Selection	Post ongoing grades and final grades	Synchronized data- post midterm and end of terms grades to DataTel. Faculty would be required to review final grades as accurate in WA and submit before being posted as the permanent grade. Ongoing synchronization of data
Registration Add/drop	Provide class roster Provide for discussion pages Provide for Instructor drop	Identification of WA as record archive
Waitlists	Add waitlist status to roster- allow for Instructor drop	
Degree Audit	Allow file sharing	CM system is main tool for faculty
Post and Archive grades	Share links and resources	Faculty would verify final grades and Positive attendance (in WA) before submission
Placement test scores	NA at this time	These functions while not available for integration at this time- are available to the student in WA
Education Plan(s)	“	
Apply for	“	
Graduation		
Payment	“	
View	“	

Documents		
Apply for	“	
Graduation		
View Financial	“	
Aid status		

What I want to suggest next is a listing of specifications that are required of this integration. I am developing this now.

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